Problem A. Balanced Eating

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

As a big eater of the Collective of Plenty, Varesa is, of course, a devoted lover of barbecue. Today, she is given n skewers, where the i-th skewer contains a_i pieces of meat.

Varesa has a peculiar eating habit: she insists on keeping her molars balanced. For each skewer, she picks a side—either left or right—at random. The first piece of meat goes to that side, the second to the other side, the third to the original side, and so on, alternating as she goes.

However, she can never remember which side she started with for any previous skewer, so she just picks randomly every time.

Varesa considers her meal to be *balanced* if the total number of pieces chewed with her left and right molars differs by at most 1. Given the number of pieces on each skewer, help her calculate the probability that her eating ends up being *balanced*. To avoid precision errors, you need to output the answer modulo 998244353.

Formally, let $M = 998244353$. It can be shown that the exact answer can be expressed as an irreducible fraction $\frac{p}{q}$, where p and q are integers and $q \not\equiv 0 \pmod{M}$. Output the integer equal to $p \cdot q^{-1} \bmod M$. In other words, output such an integer x that $0 \le x < M$ and $x \cdot q \equiv p \pmod{M}$.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains one integer n ($1 \le n \le 2 \cdot 10^5$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output one integer — the probability of being *balanced* modulo 998244353.

Example

standard input	standard output
2 1 114514 6 1 1 4 5 1 4	1 623902721

Note

In the second sample case, the answer is $\frac{3}{8}$.

Problem B. Adventure for Black

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

To make her adventure more fascinating, Kamome created a special lock for a treasure box said to contain "the island's secret." To make it challenging, she designed the lock to work with an initial sequence of numbers (let's call this sequence a). An adventurer trying to open the box would face a series of q questions, or queries, one by one. Only by answering each query correctly would the next one be revealed. If any query was answered incorrectly, the adventurer would have to start over from the very first query. These queries come in two types:

- 1 l r: Let $f(x, y) = \min\{a_x, a_{x+1}, \ldots, a_y\}$, the adventurer needs to answer $\sum_{i=l}^{r} f(i, r)$.
- 2 u: Append u to the end of a. Note that it increases the length of a by 1.

Kamome, being the creator, naturally knew all the answers. However, fast forward 10 years, and you've just found this intriguing box. Unfortunately, time has taken its toll, and Kamome has forgotten all the crucial details and answers. Your task is to help Kamome open this box!

Input

The first line of each test contains three integers n, q, and M ($1 \le n, q \le 4 \cdot 10^5$, $10^8 \le M < 10^9$).

The second line of each test contains n integers a_1, a_2, $\ldots$, a_n ($0 \le a_i < M$).

Each of the following q lines of each test contains three integers p', x, and y ($0 \le p', x, y < 2^{31}$) that describe a query. To prevent you from knowing the query parameters directly, they are given in an obfuscated form. Let Z be the answer to the most recent Type 1 query you answered (if there hasn't been a Type 1 query yet, then Z is 0). You need to compute $p = (p' + Z) \bmod 2 + 1$ to determine the type of the query:

- For Type 1 queries ($p = 1$), the actual l and r are calculated as $l = (x + Z) \bmod N + 1$ and $r = (y + Z) \bmod N + 1$, where N is the current length of a at this point. It is guaranteed that $l \le r$.
- For Type 2 queries ($p = 2$), the actual u is calculated as $u = (x + Z) \bmod M$, and the integer y has no meaning and can be ignored.

It is guaranteed that at least one of the queries is of Type 1.

Output

For each Type 1 query, output an integer — the answer to the query.

Example

standard input	standard output
4 3 100000000 2 1 2 1 0 0 2 1 99999999 2121 2 7 20	4 6

Note

In the first query, $Z = 0$, $p = (0 + 0) \bmod 2 + 1 = 1$. So $l = (0 + 0) \bmod 4 + 1 = 1$ and $r = (2 + 0) \bmod 4 + 1 = 3$. The answer is $\min\{2, 1, 2\} + \min\{1, 2\} + \min\{2\} = 1 + 1 + 2 = 4$.

In the second query, $Z = 4$, $p = (0 + 4) \bmod 2 + 1 = 2$. So $u = (99999999 + 4) \bmod M = 3$, and the array becomes $[2, 1, 2, 1, 3]$.

In the third query, $Z = 4$, $p = (2 + 4) \bmod 2 + 1 = 1$. So $l = (7 + 4) \bmod 5 + 1 = 2$ and $r = (20 + 4) \bmod 5 + 1 = 5$. The answer is $\min\{1, 2, 1, 3\} + \min\{2, 1, 3\} + \min\{1, 3\} + \min\{3\} = 1 + 1 + 1 + 3 = 6$.

Problem C. Knight

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

As Colette learns the rules of chess, she becomes intrigued by a particular piece: the knight. Its movement is notably distinct from that of other pieces. As a reminder, a knight can perform one of the following moves, with the crucial condition that the destination square must lie within the boundaries of the chessboard:

- Move k squares horizontally, and then 1 square vertically.
- Move 1 square horizontally, and then k squares vertically.

For a standard knight, $k = 2$.

Colette wants to solve the knight traversal problem for an arbitrary integer k. She considers a chessboard with n rows (numbered 1 to n) and m columns (numbered 1 to m). She places a knight, which uses this specific move parameter k, initially at the i-th row and j-th column of this chessboard. Colette's goal is to determine how many distinct squares on this $n \times m$ chessboard the knight can visit by moving any number of times (possibly zero). Can you help her?

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^5$). The description of the test cases follows.

The single line of each test case contains five integers k, n, m, i, j ($1 \le k \le 10^9$, $1 \le i \le n \le 10^9$, $1 \le j \le m \le 10^9$).

Output

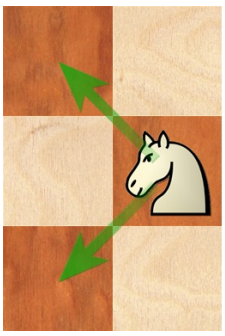
For each test case, output one integer — the number of squares that the knight can visit.

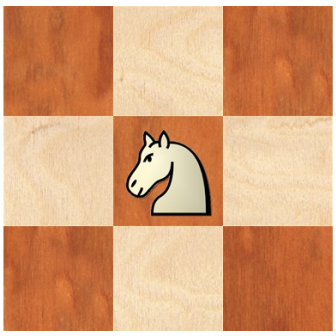
Example

standard input	standard output
4 1 3 2 2 1 2 3 3 2 2 2 3 3 1 1 3 4 5 1 2	3 1 8 9

Note

The first three example cases are illustrated in the figures below. Note that in the third figure, the variation in color is used in this instance primarily for visual clarity to help distinguish individual arrows.





Problem D. Phaethon's Melody

Input file:	`standard input`
Output file:	`standard output`
Time limit:	3 seconds
Memory limit:	256 megabytes

As Vivian heard her beloved Lord Phaethon was able to fight himself, she prepared n W-Engines and m sets of Drive Discs for him.

The i-th W-Engine provides a_i ATK. Each W-Engine also has an associated value b_i. The W-Engine's type determines to which critical hit property this bonus magnitude is applied:

- If the W-Engine is of Type 1, it contributes this $\frac{b_i}{P}$ magnitude to the total CRIT Rate.
- If the W-Engine is of Type 2, it contributes this $\frac{b_i}{P}$ magnitude to the total CRIT Damage.

The j-th set of Drive Discs obtains $\frac{x_j}{P}$ CRIT Rate and $\frac{y_j}{P}$ CRIT Damage. Phaethon can only equip one W-Engine and one set of Drive Discs at the same time, and the same property will accumulate.

When Phaethon deals damage, let his total CRIT Rate be c and his total CRIT Damage be d. These values are accumulated from the equipped W-Engine and Drive Discs. His base damage for an attack, hereafter referred to as A, is the a_i value from the currently equipped W-Engine. The damage dealt by a single attack is determined as follows:

- A critical hit occurs with a probability of $\min\{c, 1\}$. If a critical hit occurs, the damage dealt is $A \times (1 + d)$.
- If a critical hit does not occur, the damage dealt is A itself.

Since each of her n W-Engines has a distinct use under certain cases, Vivian needs to know how to choose Drive Discs to maximize the expected damage with each of the W-Engines. She wants you to help her verify her results.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 5 \cdot 10^4$). The description of the test cases follows.

The first line of each test case contains three integers n, m and P ($1 \le n, m \le 5 \cdot 10^5$, $2 \le P \le 10^9$).

Each of the following n lines contains three integers k_i, a_i and b_i ($1 \le k \le 2$, $1 \le a_i, b_i \le 10^9$), where k_i indicates the type of the i-th W-Engine.

Each of the following m lines contains two integers x_j and y_j ($1 \le x_j, y_j \le 10^9$).

Output

For each test case, output n integers $c_1, c_2, \ldots, c_n$, where c_i indicates the maximum expected damage with the i-th W-Engine. To avoid float precision error, you need to output the maximum values modulo 10^9+7. Formally, let $M = 10^9 + 7$. It can be shown that the exact answer can be expressed as an irreducible fraction $\frac{p}{q}$, where p and q are integers and $q \not\equiv 0 \pmod{M}$. Output the integer equal to $p \cdot q^{-1} \bmod M$. In other words, output such an integer x that $0 \le x < M$ and $x \cdot q \equiv p \pmod{M}$.

Please note that the modulus is different from the other problems'.

Example

standard input	standard output
1 2 2 100 1 40 80 2 40 80 60 120 80 100	88 200000099

Note

In the example, Vivian has two W-Engines and two sets of Drive Discs.

When Phaethon is equipped with the first W-Engine:

- If he is equipped with the first set of Drive Discs, he has a CRIT Rate of 140% and a CRIT Damage of 120%. Because his CRIT Rate exceeds 100%, his expected damage is $40 \times (1 + 120\%) = 88$.
- If he is equipped with the second set of Drive Discs, he has a CRIT Rate of 160% and a CRIT Damage of 100%. Because his CRIT Rate exceeds 100%, his expected damage is $40 \times (1 + 100\%) = 80$.

Therefore, the first answer is 88.

When Phaethon is equipped with the second W-Engine:

- If he is equipped with the first set of Drive Discs, he has a CRIT Rate of 60% and a CRIT Damage of 200%. He has a 60% probability to deal $40 \times (1 + 200\%) = 120$ damage, and another 40% probability to deal 40 damage, so his expected damage is $120 \times 60\% + 40 \times 40\% = 88$.
- If he is equipped with the second set of Drive Discs, he has a CRIT Rate of 80% and a CRIT Damage of 180%. He has a 80% probability to deal $40 \times (1 + 180\%) = 112$ damage, and another 20% probability to deal 40 damage, so his expected damage is $112 \times 80\% + 40 \times 20\% = 97.6$.

Therefore, the second answer is $97.6 = \frac{488}{5}$. Because $488 \cdot 5^{-1} \equiv 200000099 \pmod{10^9 + 7}$, the output is 200000099.

Problem E. LCM Queries

Input file:	`standard input`
Output file:	`standard output`
Time limit:	4 seconds
Memory limit:	256 megabytes

To test Tama's intelligence, Megumi prepares an array a of length n and q queries for her. Each query is either of the following two types:

1. Set $a_i = x$.
2. Find $d(\operatorname{lcm}(a_l, a_{l+1}, \ldots, a_r)) \bmod 998244353$. In other words, let $L = \operatorname{lcm}(a_l, a_{l+1}, \ldots, a_r)$. Tama should count the number of positive divisors of L, and then give this count modulo 998244353.

To Megumi's surprise, Tama answers her queries in 4 seconds. Given that she earns 0 points in the math test, Megumi asks you to help her verify Tama's answers. To present herself as a talented *savior*, she also wants you to calculate the answers as fast as Tama does.

Input

The first line of each test contains two integers n and q ($1 \le n, q \le 2 \cdot 10^5$).

The second line of each test contains n integers a_i ($1 \le a_i \le 10^5$).

The next q lines of each test describe the queries. Each of the lines is in either of the following two formats:

- $1\ i\ x$. ($1 \le i \le n$, $1 \le x \le 10^5$)
- $2\ l\ r$. ($1 \le l \le r \le n$)

It is guaranteed that there is at least one query of the second type.

Output

For each query of the second type, output an integer — the answer to that query.

Example

standard input	standard output
4 3 6 9 12 16 2 1 3 1 2 15 2 2 4	9 20

Note

In the first query, the 9 divisors of $\operatorname{lcm}(6, 9, 12) = 36$ are 1, 2, 3, 4, 6, 9, 12, 18, and 36.

In the second query, the 20 divisors of $\operatorname{lcm}(15, 12, 16) = 240$ are 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 16, 20, 24, 30, 40, 48, 60, 80, 120, and 240.

Problem F. Where the West Wind Ends

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

This is an interactive problem.

After Anaxagoras stated that his objective is complete, he went to meet Castorice in the nether realm. He then taught her his last lesson — the range LIS query algorithm, before he passed away. As a reminder, an LIS of a sequence is a subsequence* such that its elements are **strictly ascending** with the largest length. Castorice is very intrigued by the algorithm, and she wants to test her implementation with your help.

Specifically, she has an array a of length n, which contains integers from 1 to 3. It is also guaranteed that the array contains at least one occurrence of each integer from 1 to 3. You can ask her a query (l, r), and she gives you the length of LIS of $[a_l, a_{l+1}, \ldots, a_r]$. Your task is to determine an index i and a value x such that a_i is **NOT equal** to x.

However, since Castorice is in the nether realm, you must communicate with her through Mydei, who has limited patience. He only allows you to ask no more than 20 queries before he gets annoyed. Can you finish your task under such constraints?

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains one integer n ($3 \le n \le 10^5$).

It is guaranteed that the sum of n over all test cases does not exceed 10^5.

Interaction Protocol

To make a query, output a line in the following format (do not include quotes):

"? l r" ($1 \le l \le r \le n$)

Here, l and r represent the left and right bounds of your query.

After each query, you should read a line containing one integer, indicating the length of LIS of $[a_l, a_{l+1}, \ldots, a_r]$.

When you are ready to print the answer, output a single line in the following format:

"! i x" ($1 \le i \le n$, $1 \le x \le 3$)

Here, i represents your answer index, and x represents the value that a_i is not equal to.

You can make at most 20 queries in each test case.

The interactor is NOT adaptive, meaning that the answer is known before the participant asks the queries and doesn't depend on the queries asked by the participant.

If your program makes more than 20 queries for one test case, your program should immediately terminate to receive the verdict `Wrong Answer`. Otherwise, you can get an arbitrary verdict because your solution will continue to read from a closed stream.

After printing a query do not forget to output the end of line and flush the output. Otherwise, you may get `Idleness limit exceeded` verdict. To do this, use:

- `fflush(stdout)` or `cout.flush()` in C++;

*A sequence a is a subsequence of a sequence b if a can be obtained from b by the deletion of several (possibly, zero or all) element from arbitrary positions.

- `System.out.flush()` in Java;
- `Console.Out.Flush()` in C#;
- `stdout.flush()` in Python;
- see the documentation for other languages.

Example

standard input	standard output
1 6 3 2	 ? 1 4 ? 3 6 ! 3 1

Note

In the example, the array is $[1, 1, 2, 3, 1, 3]$.

The first query asks the length of LIS of $[1, 1, 2, 3]$, which is 3.

The second query asks the length of LIS of $[2, 3, 1, 3]$, which is 2.

Problem G. Navigation Compass

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

To collect Stellar Jade for Castorice, Soy starts to find treasures and solve puzzles in his maps. As a result, he encountered a hard puzzle named *Navigation Compass*.

An example of the *Navigation Compass*, where $a = [6, 3, 3], b = [1, 2, 4]$.

The compass is a regular m-gon, whose vertices are labeled $1, 2, \ldots, m$ in a clockwise direction, and vertex 1 is distinguished by a blue mark. The compass features n concentric rings, and these concentric rings are numbered $1, 2, \ldots, n$ from the innermost to the outermost. For each i-th ring, it has a_i *Astral Marks* and a *Celestial Axis* which initially points to vertex b_i. The quantity of *Astral Marks* determines the extent of its *Celestial Axis*'s advancement when the ring is rotated. Specifically, a single rotation causes the Celestial Axis of the i-th ring to advance a_i positions along the vertices of the m-gon in the clockwise direction.

Soy can perform n types of operations. In the i-th type of operation, Soy must rotate **simultaneously** the i-th ring and $(i \bmod n + 1)$-th ring **once**. To get the Stellar Jade, he must:

- Determine the number of distinct vertices v such that it is possible for all n *Celestial Axes* to point to the vertex simultaneously. Let this count be C. Note that he is only required to calculate C and doesn't need to perform operations actually in this subproblem.
- If $C > 0$, he should choose one of the attainable common vertices v and perform operations to make all *Celestial Axes* point to it. If multiple v exist, any one can be chosen as the goal for the sequence. Note that he doesn't need to minimize the number of operations.

Help Soy solve the puzzle and get $1,600$ Stellar Jade!

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and m ($2 \le n \le 5 \cdot 10^5$, $2 \le m \le 10^9$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i < m$).

The third line of each test case contains n integers $b_1, b_2, \ldots, b_n$ ($1 \le b_i \le m$).

It is guaranteed that the sum of n over all test cases does not exceed $5 \cdot 10^5$.

Output

For each test case, output one integer C — the number of attainable vertices.

If $C > 0$, output two extra lines:

- The first line contains one integer v — the attainable vertex Soy chooses.
- The second line contains n integers $c_1, c_2, \ldots, c_n$ ($0 \le c_i < m$), c_i indicating Soy should perform the i-th operation for c_i times. It can be shown that if a solution exists, a solution also exists within the given constraints.

Example

standard input	standard output
3 3 6 1 2 4 6 3 3 5 12 2 4 6 8 10 1 2 3 4 5 6 11 1 1 4 5 1 4 1 9 1 9 8 10	3 1 1 1 0 0 1 8 7 3 7 6 5 0

Note

In the first sample case, Soy can perform operation type 1 once and operation type 2 once, after which all *Celestial Axes* will point to vertex 1. The change of the compass is shown in the figure below.

Problem H. An AtCoder-style Problem

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

You are given a sequence of positive integers $A = (A_1, A_2, \ldots, A_N)$. For each $i = 1, 2, \ldots, N$, answer the following question:

- Determine if there exists a permutation $B = (B_1, B_2, \ldots, B_{i+M})$ of length $i + M$ and a permutation $C = (C_1, C_2, \ldots, C_i)$ of length i, such that

$$\max_{j \le k \le j+M} B_k = A_{C_j}$$

  for all $j = 1, 2, \ldots, i$.

You are given T test cases. Solve each of them.

Constraints

- All input values are integers.
- $1 \le T \le 10^4$.
- $1 \le N \le 2 \cdot 10^5$.
- $0 \le M \le 10^9$.
- $1 \le A_i \le N + M$.
- The sum of N over all test cases doesn't exceed $2 \cdot 10^5$.

Input

Input is given from Standard Input in the following format:

T
case_1
case_2
$\vdots$
case_T

case_k denotes the k-th test case. Each test case is given in the following format:

$N \quad M$
$A_1 \; A_2 \; \ldots \; A_N$

Output

Output T lines. The k-th line ($1 \le k \le T$) should contain the answer for the k-th test case.

For each test case, output a binary string S of length N. The i-th character should be `1` if such B and C exist for prefix $(A_1, A_2, \ldots, A_i)$, and `0` if it doesn't exist.

Example

standard input	standard output
3 4 1 3 3 5 5 4 2 3 3 5 5 6 1 6 5 4 3 2 1	0101 1000 000010

Note

In the first sample case:

- For $A = (3, 3)$, we can construct $B = (2, 3, 1)$ and $C = (1, 2)$.
- For $A = (3, 3, 5, 5)$, we can construct $B = (4, 5, 1, 3, 2)$ and $C = (3, 4, 2, 1)$.

Problem I. FST: First Search Traversal

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

As her assignment, Riri's given two permutations[†] of length n. Determine if there exists a rooted tree such that the first permutation is one of its DFS orders and the second permutation is one of its BFS orders.

As a reminder, the DFS order is generated by the following pseudocode:

```
dfs_order = []

def dfs(v):
    dfs_order.append(v)
    pick an arbitrary permutation s of children of v
    for child in s:
        dfs(child)

dfs(root)
```

and the BFS order is generated by the following pseudocode:

```
bfs_order = []
queue = [root]
while len(queue):
    u = queue[0]
    bfs_order.append(u)
    queue = queue[1:]
    pick an arbitrary permutation s of children of u
    for child in s:
        queue.append(child)
```

However, Riri is addicted to kemurikusa and is not interested in these boring assignment, so she forwards the assignment to you. Can you do it well?

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains one integer n ($1 \le n \le 2 \cdot 10^5$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le n$, $a_i \ne a_j$) — the first permutation.

The third line of each test case contains n integers $b_1, b_2, \ldots, b_n$ ($1 \le b_i \le n$, $b_i \ne b_j$) — the second permutation.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output "`YES`" if such a tree exists; otherwise, output "`NO`". You can output the answer in any case (upper or lower). For example, the strings "`yEs`", "`yes`", "`Yes`", and "`YES`" will be recognized as positive responses.

[†] A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, $[2, 3, 1, 5, 4]$ is a permutation, but $[1, 2, 2]$ is not a permutation (2 appears twice in the array), and $[1, 3, 4]$ is also not a permutation ($n = 3$ but there is 4 in the array).

Example

standard input	standard output
2 2 1 2 1 2 2 1 2 2 1	YES NO

Problem J. k-MEX

Input file:	`standard input`
Output file:	`standard output`
Time limit:	5 seconds
Memory limit:	256 megabytes

Kagari is managing a finished world tree. The tree is rooted at a specific vertex r, and each vertex has an integer value. To process the tree more efficiently, she may do the following operation on an arbitrary vertex v ($v \neq r$):

1. Let the path from vertex r to vertex v consist of the sequence of vertices $u_1, u_2, \ldots, u_m$, where $u_1 = r$ and $u_m = v$.
2. Remove the edges $(u_1, u_2), (u_2, u_3), \ldots, (u_{m-1}, u_m)$.
3. Connect vertex r with $u_2, u_3, \ldots, u_m$ directly.

During this process, Kagari may also query the k-MEX[‡] value of the set of vertex integer values in a given subtree[§]. Can you help her program to answer her queries efficiently?

Input

The first line of each test contains three integers n, q, and r ($1 \leq n, q \leq 10^6$, $1 \leq r \leq n$) — the size of the tree, the number of queries and operations, and the root vertex.

The second line of each test contains n integers $a_1, a_2, \ldots, a_n$ ($0 \leq a_i \leq n$) — the integer values of each vertex.

The following $n - 1$ lines of each test describe the tree. Each of the lines contains two integers u and v that indicate an edge between vertex u and v. It is guaranteed that these edges form a tree.

The following q lines of each test describe the queries and operations. Each of the lines is either of:

- $1\ v$ ($1 \leq v \leq n, v \neq r$) that indicates an operation on vertex v.
- $2\ v\ k$ ($1 \leq v \leq n$, $1 \leq k \leq 10$) that indicates a k-MEX query of the subtree of vertex v.

Output

For each k-MEX query, output an integer — the answer to that query.

[‡]The k-th minimum excluded (k-MEX) of a collection of integers $c_1, c_2, \ldots, c_n$ is defined as the k-th smallest non-negative integer x which does not occur in the collection c.

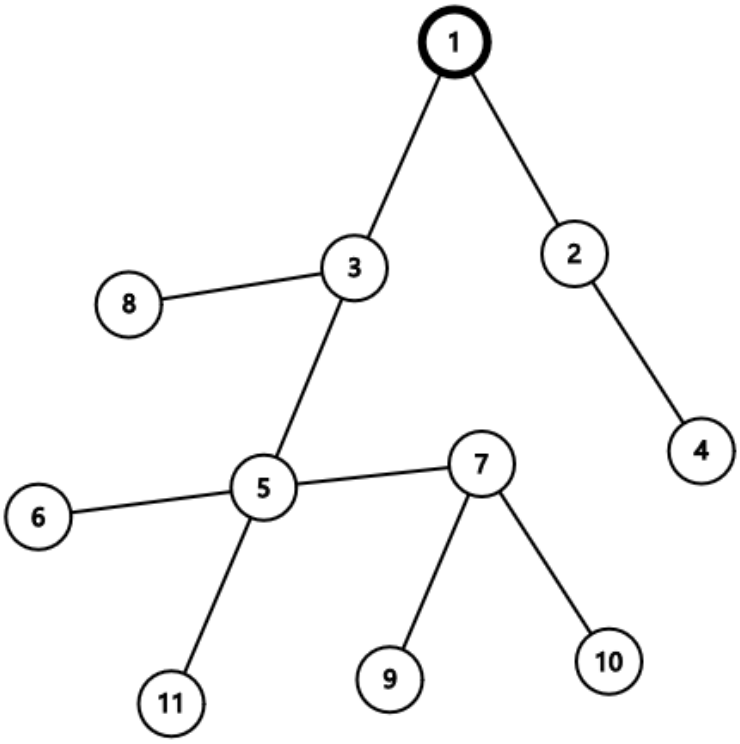
[§]A vertex u is said to be in the subtree of vertex v if and only if the path from u to the root contains v.

Example

standard input	standard output
11 6 1 4 5 6 0 2 1 8 2 3 0 4 1 2 1 3 2 4 3 5 3 8 5 6 5 7 7 9 7 10 5 11 2 3 1 1 7 2 5 3 2 4 2 1 9 2 1 2	5 5 2 9

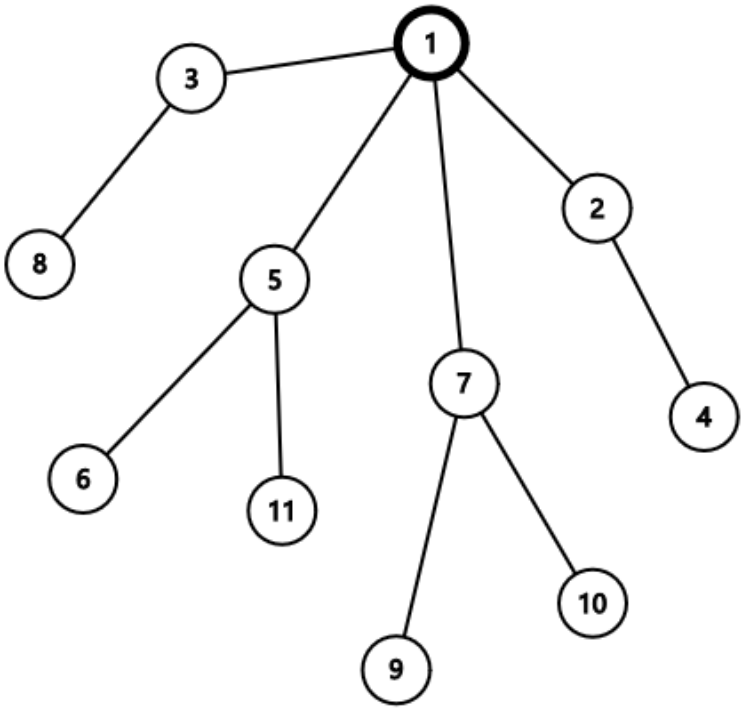
Note

The initial tree is shown in the figure below.



In the first query, the values of the subtree of vertex 3 are $[6, 2, 1, 8, 2, 3, 0, 4]$, and its 1-MEX is 5.

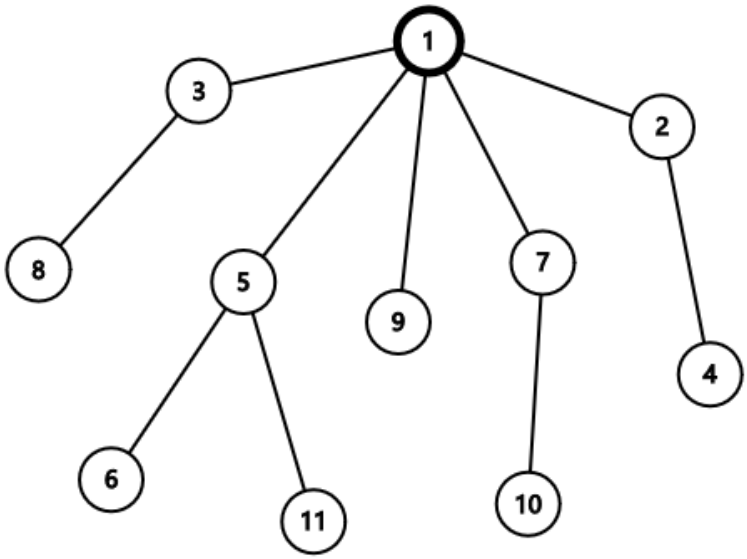
After the operation on vertex 7, the tree is shown in the figure below.



In the second query, the values of the subtree of vertex 5 are $[2, 1, 4]$, and its 3-MEX is 5.

In the third query, the values of the subtree of vertex 4 are $[0]$, and its 2-MEX is 2.

After the operation on vertex 9, the tree is shown in the figure below.



In the fourth query, the values of the subtree of vertex 1 are $[4, 5, 6, 0, 2, 1, 8, 2, 3, 0, 4]$, and its 2-MEX is 9.

Problem K. Pleasure of Hope

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

> 'Tis distance lends enchantment to the view, and robes the mountain in its azure hue.
> Thus, with delight, we **linger** to survey the promised joys of life's unmeasured way.
> Thus, from afar, each dim-discovered scene more pleasing seems than all the past hath been,
> And every form, that Fancy can repair from dark oblivion, glows divinely there.
>
> Thomas Campbell, *Pleasure of Hope*

You are given an integer n and q pairs of integers (l_j, r_j). Construct a sequence a such that:

1. a contains n **distinct** positive integers.
2. For all $1 \le i < n$, $\gcd(a_i, a_{i+1}) = 1$.
3. For all $1 \le j \le q$, $\sum_{k=l_j}^{r_j} a_k$ is a composite number[¶].

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 200$). The description of the test cases follows.

The first line of each test case contains two integers n and q ($1 \le q \le n \le 4 \cdot 10^4$).

Each of the following q lines of each test case contains two integers l_j and r_j ($1 \le l_j \le r_j \le n$).

It is guaranteed that the sum of n over all test cases does not exceed $4 \cdot 10^4$.

Output

For each test case, output n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^{18}$, $\sum_{i=1}^{n} a_i \le 10^{18}$). It can be shown that a solution always exists under the given constraints.

Example

standard input	standard output
2 1 1 1 1 4 2 1 3 2 4	114514 1 2 3 4

[¶]An integer n is a composite number if and only if it has at least three distinct divisors.

Problem L. AL-1S

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	256 megabytes

The name of Tendou "Alice" Aris originates from the misinterpretation of her identifier `AL-1S`, when Saiba Momoi misread the digit `1` as the uppercase letter `I`. This confusion arises because the characters `l` (lowercase L), `I` (uppercase i), and `1` (digit one) look annoyingly similar. Unfortunately, there's no way to prevent all three—`l`, `I`, and `1`—from appearing together in the same string. That's why her maid Key decided to lay down some strict rules to avoid potential misinterpretation.

A string s is said to be *approved* by Key if it satisfies all of the following:

- `l`, `I`, and `1` must not appear at the beginning or end of the string.
- Every digit `1` can be only adjacent to digits.
- Every letter `l` can be only adjacent to lowercase letters.
- Every letter `I` can be only adjacent to uppercase letters.

For example:

- "`abclllz2113AabCIIXYZ`" is *approved*.
- "`1abcd`", "`AbcllZ`", and "`ABCDI`" are *disapproved*.

You're given a string s that may contain some `?` characters. You need to replace each `?` with a Latin letter or digit so that the resulting string is approved by Key. Count the number of valid completions. Since the result can be too large, you only need to answer the result modulo 998244353.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The single line of each test case contains one string s ($1 \le |s| \le 10^6$). It is guaranteed that s contains only Latin characters, digits, and question marks.

It is guaranteed that the sum of $|s|$ over all test cases does not exceed 10^6.

Output

For each test case, output one integer — the number of strings *approved* by Key modulo 998244353.

Example

standard input	standard output
4 ? ??? ai???IZ ??I	59 206710 89775 0

Problem M. Bridge IV

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Since they cannot find the third and fourth players, Diluc and Kaeya are playing a simplified version of contract bridge with only two players. A full deck of 52 cards is evenly split into two visible hands. Each player is randomly and independently marked as **vulnerable** or not.

Then, they start to bid the contract in turn, with Diluc bidding first. Players bid in turn, starting with Diluc. A bid is a contract consisting of a level (1–7) and a suit from the following hierarchy: `C < D < H < S < N` (No Trump). A bid can only be made if it *outranks* the current highest bid — that is, it has a higher level, or the same level with a higher suit. For example, if Diluc has bid `1H`, Kaeya can bid `1S`, `1N`, or `2C`, but he cannot bid `1C` or `1H` itself. Either player may pass. If both pass consecutively, bidding ends. The last bidder becomes the **declarer**, and the other becomes the **defender**.

The declarer must win at least $x + 6$ tricks, where x is the contract level. After all cards are played, the score is calculated. The declarer gains points if the contract is fulfilled; otherwise, the defender gains points. The total score is always zero-sum.

Let x be the contract level, and y be the number of extra tricks ($y > 0$ if over, $y < 0$ if under):

- If the declarer fulfills the contract, his score consists of four parts:
  1. **Trick Points**:
     - $20x$ for suits `C` or `D`
     - $30x$ for suits `H` or `S`
     - $30x + 10$ for suit `N`
  2. **Contract Bonus**:
     - If Trick Points ≥ 100: $+500$ if vulnerable, $+300$ otherwise
     - Otherwise: $+50$
  3. **Overtrick Bonus** (if $y > 0$):
     - $20y$ for suits `C` or `D`
     - $30y$ for suits `H`, `S`, or `N`
  4. **Slam Bonus**:
     - If $x = 7$: $+1500$ if vulnerable, $+1000$ otherwise
     - If $x = 6$: $+750$ if vulnerable, $+500$ otherwise
- If the contract fails ($y < 0$), let $z = -y$ be the number of undertricks. The defender will gain score:
  - If the declarer is vulnerable:
    - $z = 1, 2, 3$: 200, 500, 800
    - $z \geq 4$: $300z - 100$
  - If the declarer is not vulnerable:
    - $z = 1, 2, 3$: 100, 300, 500
    - $z \geq 4$: $300z - 400$

For example, if Diluc is not vulnerable, declares `6N` and wins 13 tricks, he will win $(30 \times 6 + 10) + 300 + 30 + 500 = 1020$ points. If Kaeya is vulnerable, declares `4S` and wins 4 tricks, he will lose $300 \times (10 - 4) - 100 = 1700$ points.

Specifically, if neither bids a contract during the bidding, neither of them gains score.

Both players are perfect analysts: they know exactly how many tricks they can win as declarer with each suit. Given this, bidding proceeds optimally for both sides to maximize their expected score. They are now tired of playing cards and want to know their score as long as the cards are assigned. Because double-dummy analysis is very complex, they will tell you the results and expect you to calculate the final score within 1 second.

Input

Each test contains multiple test cases. The first line contains one integer t ($1 \le t \le 10^4$) — the number of test cases. The description of each test case follows.

The first line of each test case contains two integers v_1, v_2 ($0 \le v_1, v_2 \le 1$) — vulnerability of Diluc and Kaeya (1 means vulnerable).

The second line of each test case contains five integers c_1, d_1, h_1, s_1, n_1 ($0 \le c_1, d_1, h_1, s_1, n_1 \le 13$) — the number of tricks Diluc can win as declarer with each suit `C`, `D`, `H`, `S`, and `N`.

The third line of each test case contains five integers c_2, d_2, h_2, s_2, n_2 ($0 \le c_2, d_2, h_2, s_2, n_2 \le 13$) — the same numbers for Kaeya.

It is guaranteed that each test case corresponds to a valid double dummy analysis with a consistent deal.

Output

For each test case, output one integer — the final score Diluc gets, assuming both play optimally.

Example

standard input	standard output
6 0 0 9 11 13 6 9 3 2 0 7 3 0 1 9 11 13 6 9 3 2 0 7 3 0 0 9 6 7 3 5 4 6 5 10 6 1 0 4 5 2 4 2 9 8 11 9 11 0 1 5 5 7 6 5 8 8 6 6 5 0 1 10 7 9 5 5 3 6 3 8 8	1400 1510 -300 -460 -90 140

Note

In the first sample case, the optimal contract is `7S` declared by Kaeya.

In the second sample case, the optimal contract is `7H` declared by Diluc.